PROJECT SYNOPSIS

ON

LostLink: An Intelligent Lost-and-Found System with Multi-Modal Matching and Evidence-Based Ownership Verification

Submitted to the Department of Computer Science and Engineering
in Partial Fulfillment of the Requirements
for the Degree of

Bachelor of Technology

in

Computer Science and Engineering

by

Rohit Kumar (2300910100156)
Yuvraj Mishra (2300910100219)
Aman Saleem (2300910100027)

Under the Supervision of

Ms. Priya Singh
Department of Computer Science and Engineering
JSS Academy of Technical Education, Noida



JSS ACADEMY OF TECHNICAL EDUCATION, NOIDA
DR. APJ ABDUL KALAM TECHNICAL UNIVERSITY
UTTAR PRADESH, LUCKNOW
(Formerly Uttar Pradesh Technical University, Lucknow)
SEPTEMBER, 2026

Vision and Mission

Vision of the Institute

JSS Academy of Technical Education Noida aims to become an Institution of excellence in imparting quality Outcome Based Education that empowers the young generation with Knowledge, Skills, Research, Aptitude and Ethical values to solve Contemporary Challenging Problems.

Mission of the Institute

1. Develop a platform for achieving globally acceptable level of intellectual acumen and technological competence.
2. Create an inspiring ambience that raises the motivation level for conducting quality research.
3. Provide an environment for acquiring ethical values and positive attitude.

Vision of the Department

"To spark the imagination of the Computer Science Engineers with values, skills and creativity to solve the real-world problems."

Mission of the Department

1. To inculcate creative thinking and problem-solving skills through effective teaching, learning and research.
2. To empower professionals with core competency in the field of Computer Science and Engineering.
3. To foster independent and lifelong learning with ethical and social responsibilities.

Program Outcomes (POs)

Engineering Graduates will be able to:

PO1: Engineering knowledge

Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2: Problem analysis

Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3: Design/development of solutions

Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4: Conduct investigations of complex problems

Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5: Modern tool usage

Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO6: The engineer and society

Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7: Environment and sustainability

Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO8: Ethics

Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9: Individual and teamwork

Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10: Communication

Communicate effectively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11: Project management and finance

Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12: Life-long learning

Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Program Educational Outcomes (PEOs)

PEO1 To apply computational skills necessary to analyze, formulate and solve engineering problems.

PEO2 To establish as entrepreneurs, and work in interdisciplinary research and development organizations as an individual or in a team.

PEO3 To inculcate ethical values and leadership qualities in students to have a successful career.

PEO4 To develop analytical thinking that helps them to comprehend and solve real-world problems and inherit the attitude of lifelong learning for pursuing higher education.

Program Specific Outcomes (PSOs)

PSO1 Acquiring in-depth knowledge of theoretical foundations and issues in Computer Science to induce learning abilities for developing computational skills.

PSO2 Ability to analyse, design, develop, test and manage complex software system and applications using advanced tools and techniques.

Course Outcomes and CO–PO–PSO Mapping

Course Outcomes (COs)

C410.1 Identify, formulate, design and analyze a research based/web based problem.

C410.2 Communicate effectively in verbal and written form.

C410.3 Apply appropriate computing and engineering skills for obtaining solution to the formulated problem within a stipulated time.

C410.4 Work effectively as a part of team in multi-disciplinary areas.

C410.5 Consolidate the final outcome in the form of a publication.

CO–PO–PSO Mapping

Table 1: CO–PO–PSO mapping matrix (correlation levels 1–3)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
C410.1	3	3	3	3	2	3	3	3	3	3	2	3	3	3
C410.2	2	2	2	2	2	2	2	2	2	3	2	3	2	3
C410.3	3	3	3	3	3	3	2	3	3	3	3	3	3	3
C410.4	3	3	3	3	2	3	2	3	3	3	3	3	3	3
C410.5	3	3	3	3	3	3	3	3	3	3	3	3	3	3
C410	2.80	2.80	2.80	2.80	2.40	2.80	2.40	2.80	2.80	3.00	2.60	3.00	2.80	3.00

Declaration

We hereby declare that this synopsis submission is our own work and that, to the best of our knowledge and belief, it contains no material previously published or written by another person nor material which to a substantial extent has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text.

Name: Rohit Kumar
Roll No.: 2300910100156 _____

(Candidate Signature)

Name: Yuvraj Mishra
Roll No.: 2300910100219 _____

(Candidate Signature)

Name: Aman Saleem
Roll No.: 2300910100027 _____

(Candidate Signature)

Certificate

This is to certify that the project synopsis report entitled **"LostLink: An Intelligent Lost-and-Found System with Multi-Modal Matching and Evidence-Based Ownership Verification"** which is submitted by Rohit Kumar, Yuvraj Mishra and Aman Saleem in partial fulfillment of the requirement for the award of degree B.Tech. in the Department of Computer Science and Engineering of Dr. APJ Abdul Kalam Technical University, Uttar Pradesh, Lucknow is a record of the candidates' own work carried out by them under my supervision. The matter embodied in this report is original and has not been submitted for the award of any other degree.

Signature

Ms. Priya Singh

Designation: Assistant Professor

Address: JSS Academy of Technical Education, Noida

Date: _____

Abstract

Lost-and-found recovery on campuses today relies almost entirely on notice boards, informal messaging groups, and keyword-driven listing portals, which struggle to surface the single genuine match among many similar-looking reports and offer no reliable way to confirm that a claimant is the true owner of a recovered item. **LostLink** is proposed as a web-based platform that addresses both halves of this problem. On the matching side, it uses zero-shot, similarity-native visual embedding models (CLIP and DINOv2) rather than category-specific fine-tuned networks, so that image similarity can be evaluated across a broad range of item categories without collecting a fresh labelled dataset for each one; this visual signal is fused with text, location, and time similarity into a single weighted, explainable match score. On the verification side, LostLink introduces a scored, evidence-based ownership-verification mechanism in which a claimant must supply private identifying details not shown on the public listing, evaluated using the False Acceptance Rate / False Rejection Rate framework standard in authentication research, with multi-claimant disputes resolved by ranking. Matching performance will be evaluated using precision, recall, F1-score, and top-K retrieval accuracy against a keyword/category baseline; verification performance will be evaluated using FAR/FRR across thresholds and identification-rate-at-rank for contested claims. The project targets a centralised, transparent, and auditable recovery workflow, closing two gaps identified in the existing literature: the absence of a cross-category zero-shot matching evaluation, and the absence of a quantitatively measured ownership-verification mechanism for lost-and-found claims.

List of Figures

2.1 The five-stage LostLink pipeline, from report submission to a decision on a claim. 4

5.1 LostLink system architecture / block diagram. 10

6.1 Project timeline (Gantt chart), August 2026 – January 2027. 11

List of Tables

1 CO–PO–PSO mapping matrix (correlation levels 1–3) v

4.1 Comparison of existing lost-and-found and image-matching systems . . . 6

5.1 Technologies and tools used 9

6.1 Project phases and their planned months 12

List of Symbols and Abbreviations

Symbol/Abbreviation	Description
CLIP	Contrastive Language–Image Pre-training
DINOv2	Self-Distillation with No Labels, version 2
CNN	Convolutional Neural Network
FAR	False Acceptance Rate
FRR	False Rejection Rate
EER	Equal Error Rate
API	Application Programming Interface
JWT	JSON Web Token
TF-IDF	Term Frequency–Inverse Document Frequency
UI/UX	User Interface / User Experience
DB	Database
ML	Machine Learning
RQ	Research Question

Contents

Vision and Mission i

Program Outcomes, PEOs and PSOs ii

Course Outcomes and CO–PO–PSO Mapping v

Declaration vi

Certificate vii

Abstract viii

List of Figures ix

List of Tables x

List of Symbols and Abbreviations xi

1 Introduction 1

1.1 Problem Statement 1

1.2 Motivation 2

2 Objective(s) 3

2.1 Primary Objectives 3

2.2 Research Objectives 3

2.3 Proposed Pipeline 3

3 Scope of the Project 5

3.1 In Scope 5

3.2 Proposed System Modules 5

4 Related Work 6

4.1 Conclusion 8
4.2 Future Scope 8

5 Hardware and Software Requirements 9
5.1 Software Requirements 9
5.2 Hardware Requirements 9
5.3 System Architecture 10

6 Predicted Timeline for the Project 11

7 Conclusion 13

References 14

A Plagiarism Report 16
A.1 Plagiarism Report 16
A.2 AI Content Report 16

Chapter 1

Introduction

Personal items get lost every day in colleges, offices and public places. People today rely completely on notice boards, social media groups and talking to others to find their things. Current digital systems for lost and found items mostly show a list that people can search through.. Finding the right match among many similar entries is really hard.. Proving that the person who found something is actually the real owner is even more difficult.

LostLink is a system that uses images, text, location and time to find matches automatically. It also has a step to check if the person claiming an item is really the owner. This system is meant to save time and effort when trying to get lost things back. It also helps make the whole process clear and easy to track for people, in charge.

1.1 Problem Statement

Several distinct difficulties make lost-and-found recovery unreliable in practice:

- It is hard to find lost or found reports because there are so many listings.
- Searching with keywords misses items that look the same but are described with words.
- Two people often describe the object in very different ways.
- Existing systems collect information, about location and time. Do not use it much when matching items.
- Many people might say they found the item but there is no good way to check their claims.
- Checking if someone really owns an item is slow not the every time and depends completely on what an administrator thinks.

This makes me think about the question: how can an intelligent system find possible matches and check if they are real while making fewer wrong and fake claims?

1.2 Motivation

- I think we should cut the time needed to retrieve lost belongings.
- I think we should cut reliance on manual searching in notice boards and social media groups.
- I think we should boost matching accuracy by combining image, text, location and time data than using only keywords.
- I think we should lower incorrect ownership claims by using a structured verification process.
- I think we should offer a centralized recovery workflow instead of scattered informal channels.
- I think we should keep the recovery process transparent and auditable so administrators can see why a match or claim was accepted.

Chapter 2

Objective(s)

2.1 Primary Objectives

- Build a centralised lost-and-found platform that everyone can use.
- Automatically show matches when new lost items are reported.
- Employ pretrained computer-vision models to find images.
- Merge visual clues with signals such, as text, location and time.
- Create a scoring system that verifies ownership of items.
- Cut down on fraudulent claims.

2.2 Research Objectives

RQ1 — How do zero-shot embeddings like CLIP and DINOv2 compare to trained and category-specific fusion models when it comes to -category lost-and-found matching? **RQ2** — Can the false-acceptance rate for fraudulent claims be quantitatively reduced by a scored, evidence-based verification system without significantly increasing the false rejections of legitimate owners?

2.3 Proposed Pipeline

Figure 2.1 summarises the five-stage pipeline through which every report and claim passes.

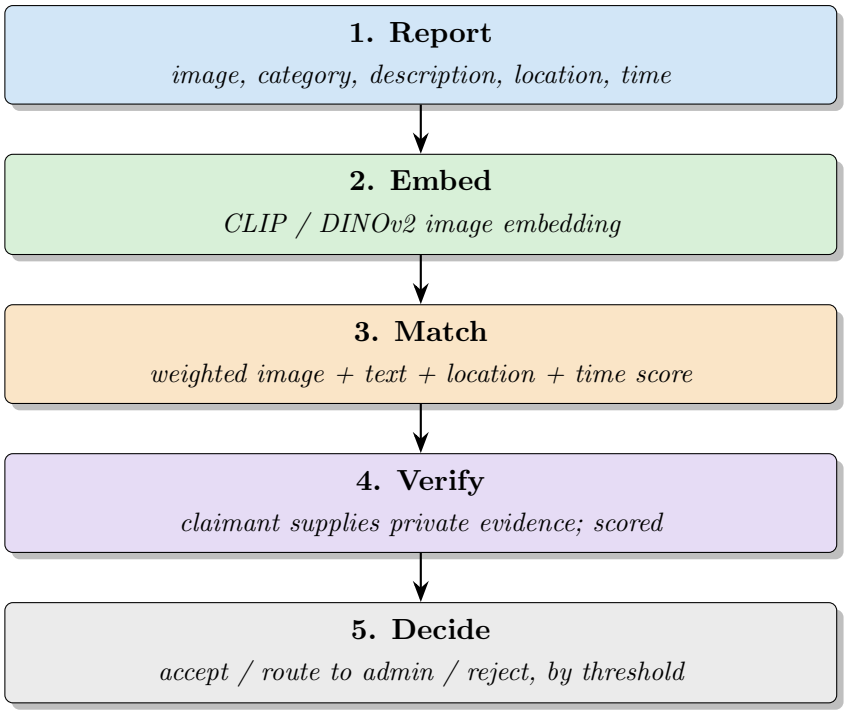


Figure 2.1: The five-stage LostLink pipeline, from report submission to a decision on a claim.

Chapter 3

Scope of the Project

3.1 In Scope

- Web application front end (React).
- Lost and found reporting with image upload.
- Image similarity using pretrained CLIP / DINOv2 embeddings.
- Text, location, and time similarity.
- Multi-factor match-score generation.
- Ownership verification and scoring.
- Multiple-claim ranking.
- Notifications for potential matches and claim status.
- Administrator dashboard for review and approval.
- Recovery history and audit trail.

3.2 Proposed System Modules

User registration and authentication, lost-item reporting, found-item reporting, image embedding generation (CLIP/DINOv2), text similarity analysis, location and time matching, multi-factor match-score generation, potential-match notification, ownership verification, claim management and ranking, administrator verification dashboard, secure handover, and recovery history are the modules that make up the system.

Chapter 4

Related Work

Several pertinent systems already exist, but they only partially address the issue LostLink is intended to solve, according to a review of recent lost-and-found and image-matching literature.

Table 4.1: Comparison of existing lost-and-found and image-matching systems

System	What it does	Where it falls short
Dhanawardhana et al. (2025) [1]	MobileNetV2-based Siamese network, fine-tuned on 848 bag-image pairs; fuses image, text, location and time similarity into one score.	Fine-tuning ties the model to a single item category (bags); no ownership-verification component.
LostNet (Zhou et al., 2024) [2]	MobileNetV2 with CBAM attention, fine-tuned for image matching, combined with perceptual hashing.	Image-only; single-category focus; no text, location, time, or verification signal.
Campus claim-arbitration system (2026, preprint) [6]	Describes an ownership-verification and multi-claim arbitration workflow for campus lost-and-found.	Non-peer-reviewed; no quantified fraud-detection or verification-accuracy metrics published.
Typical campus lost-and-found portals [7, 8]	Registration, keyword/category search, notifications, and an administrator panel.	Matching is keyword or category based; the correct item is easy to miss among similar listings.

Keyword- and metadata-based matching. Registration, a keyword or category search form, and an administrator panel that manually reviews reports are the foundation of the majority of implemented lost-and-found portals, including standard campus systems [8, 7]. On top of this framework, UniFind applies natural language processing and basic image recognition; however, instead of using evidence related to the item itself, it verifies claimants using college login credentials [7]. While blockchain-based proposals concentrate on a tamper-resistant audit trail while leaving matching to keyword search [17], map-based systems add location context to listings while matching is still dependent on manually entered fields [16]. Keyword matching misses true matches whenever descriptions differ because two people seldom describe the same thing using the same words.

Fine-tuned convolutional and Siamese matching. Most category-specific lost-and-found matchers are based on the Siamese-network concept, which was first presented by Koch et al. for one-shot image recognition [10]. Using 848 hand-labeled bag-image

pairs, Dhanawardhana et al. optimize a MobileNetV2-based [13] Siamese network and combine the final score with text, location, and time similarity [1]. LostNet reports 96.8% test accuracy [2], combining attention and perceptual hashing with a MobileNetV2 backbone. Only about 30% accuracy is reported in a different ResNet-50-based similarity study [11], demonstrating that a classification-oriented backbone does not automatically transfer to open-set similarity matching. The common limitation is that a Siamese network's utility is solely dependent on the category on which it was optimized; expanding it to a different category necessitates a new labeled dataset.

Zero-shot vision-language and self-supervised embeddings. CLIP is trained using image and caption pairs. It makes sure that similar images and text are close to each other in a shared space. It works well without needing special adjustments for tasks. A paper from 2021 talks about this. DINOv2 uses a method. It uses self-supervised learning to create features. It does not need labels. A paper from 2022 explains this. The two methods are not the same. Encoders that use self-supervised learning like DINOv2 often separate images from the same category better. CLIP is better when text helps match images. Two papers from 2026 and 2024 show this. A system based on CLIP called ViT-L/14 got 95.2 percent accuracy. It matched images to text on a group of 500 images from CIFAR-10. A paper from 2025 mentions this. This is much better than systems that use CNNs.. This is on a general test not on images that are almost the same and from one category. These are the kinds of images found in lost-and-found reports. No system, in the review compared CLIP and DINOv2 directly for lost-and-found images.

Multimodal fusion. Even though two different black backpacks may appear almost identical, location and time context can distinguish them, so image similarity alone is insufficient. It has been demonstrated that fusing features across multiple encoder layers, rather than just the final embedding, improves cross-modal retrieval [12], and general strategies for combining image and text embeddings are well studied [5]. Although the per-modality contribution is rarely isolated and measured, Dhanawardhana et al.'s weighted fusion of image, text, location, and time similarity is the clearest precedent within lost-and-found specifically [1].

Ownership verification. The majority of current prototypes do not confirm that the claimant is the actual owner once a candidate match has been found. A 2026 preprint describes a claim-arbitration workflow but reports no measured accuracy [6]; a broader survey attributes a significant portion of never-recovered campus items to the lack of a trustworthy ownership check [8]; and UniFind verifies claimants using college-ID login rather than item-specific evidence [7]. A well-researched authentication problem outside of this literature is confirming a claimed identity from provided evidence. This is assessed using False Acceptance Rate (FAR) and False Rejection Rate (FRR) across thresholds [14]. Benchmarks like EVI apply this to knowledge-based verification and

report identification-rate-at-rank for multi-claimant scenarios [15].

4.1 Conclusion

This review highlights two gaps. The question of whether zero-shot embeddings, and DINOv2 in particular, can match or surpass this accuracy across a wide range of item categories *without* any fine-tuning is still open and testable. First, matching systems that report strong accuracy achieve it by fine-tuning on a single object category [1, 2]. While a standard, threshold-based FAR/FRR evaluation framework already exists in the authentication literature and has just not been applied here [14, 15], ownership verification in lost-and-found systems is currently either absent, reduced to a login check, or described only as a workflow without measured performance [7, 6]. Instead of a more descriptive prototype, LostLink is intended to bridge both gaps with quantifiable, cross-category outcomes.

4.2 Future Scope

- Mobile application.
- QR-code-based item registration.
- Integration with campus security systems.
- A cross-campus recovery network.
- Fine-tuning of embedding models if a sufficiently large in-domain dataset becomes available.

Chapter 5

Hardware and Software Requirements

5.1 Software Requirements

Table 5.1: Technologies and tools used

Category	**Tools / Technologies**
Frontend	React.js, Tailwind CSS
Backend	Node.js, Express.js
Database	MongoDB
ML Service	Python
Image Processing	OpenCV
Image Embeddings	CLIP, DINOv2
Similarity Computation	Cosine similarity
Text Matching	TF-IDF / sentence embeddings
Maps	Leaflet + OpenStreetMap, or Google Maps API
Authentication	JWT
Notifications	Firebase Cloud Messaging, Email
Version Control / Deployment	Git, GitHub; Vercel (frontend); Render/Railway (backend); MongoDB Atlas

5.2 Hardware Requirements

- **Development machine:** For frontend and backend development, a laptop or desktop with at least 8 GB RAM and a contemporary multi-core CPU is adequate; for quicker local experimentation with CLIP/DINOv2 embedding generation, a CUDA-capable GPU (such as 6 GB+ VRAM) is advised but not necessary.
- **Inference/deployment:** For the anticipated report volume of a campus-scale deployment, embedding generation can run on CPU; if report volume increases, a small GPU-backed instance (such as a free or low-tier cloud GPU) would lower latency.
- **Storage:** Use cloud object storage or the deployment platforms file storage for storing uploaded item photographs. For data, during development MongoDB Atlas free tier is sufficient.

- **Network:** I need an internet connection for API calls between the frontend, backend and the Python ML microservice and for third-party services such, as maps and notifications.

5.3 System Architecture

The layered architecture is depicted in Figure 5.1, where the client application interacts with a Node.js API that assigns the task of computing image and text similarity to a specialized Python machine-learning service.

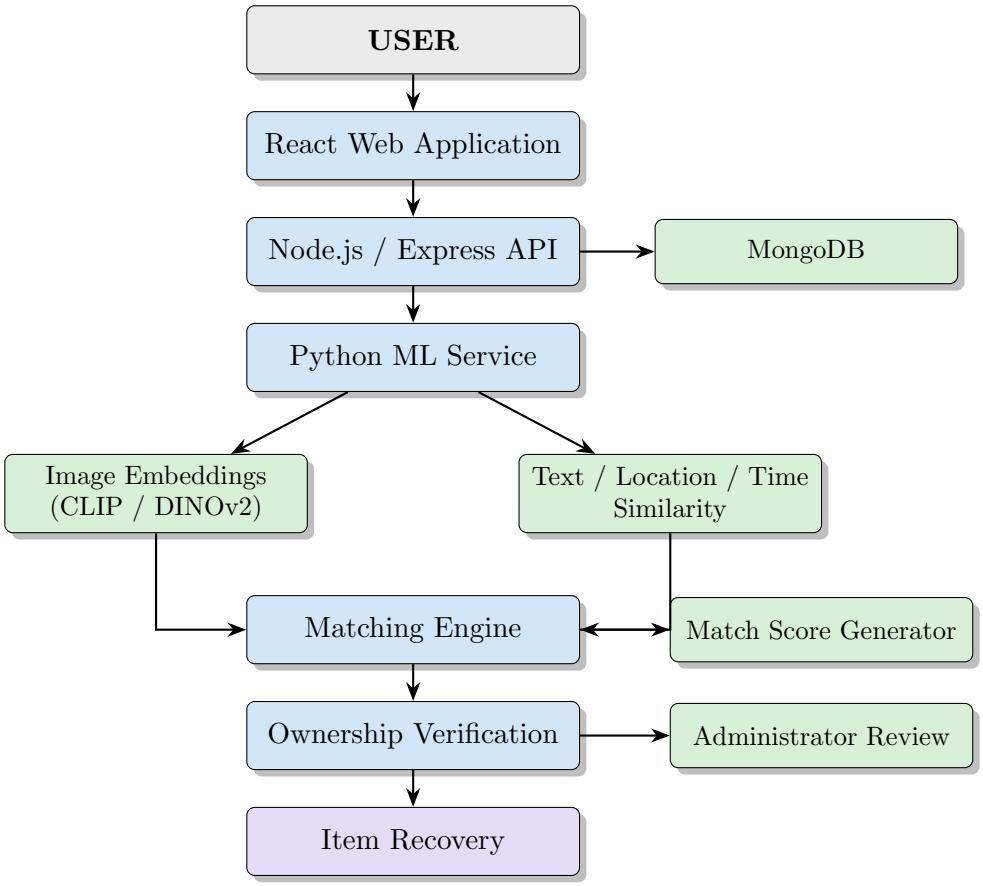


Figure 5.1: LostLink system architecture / block diagram.

The core matching computation combines four weighted similarity signals into a single score:

$$S = w_1 S_{\text{image}} + w_2 S_{\text{text}} + w_3 S_{\text{location}} + w_4 S_{\text{time}}, \qquad \sum_i w_i = 1 \tag{5.1}$$

with the weights w_i tuned on a held-out validation split rather than fixed a priori, following the fusion structure used by Dhanawardhana et al. [1].

Chapter 6

Predicted Timeline for the Project

The project is planned over a six-month window from August 2026 to January 2027, as shown in Figure 6.1.

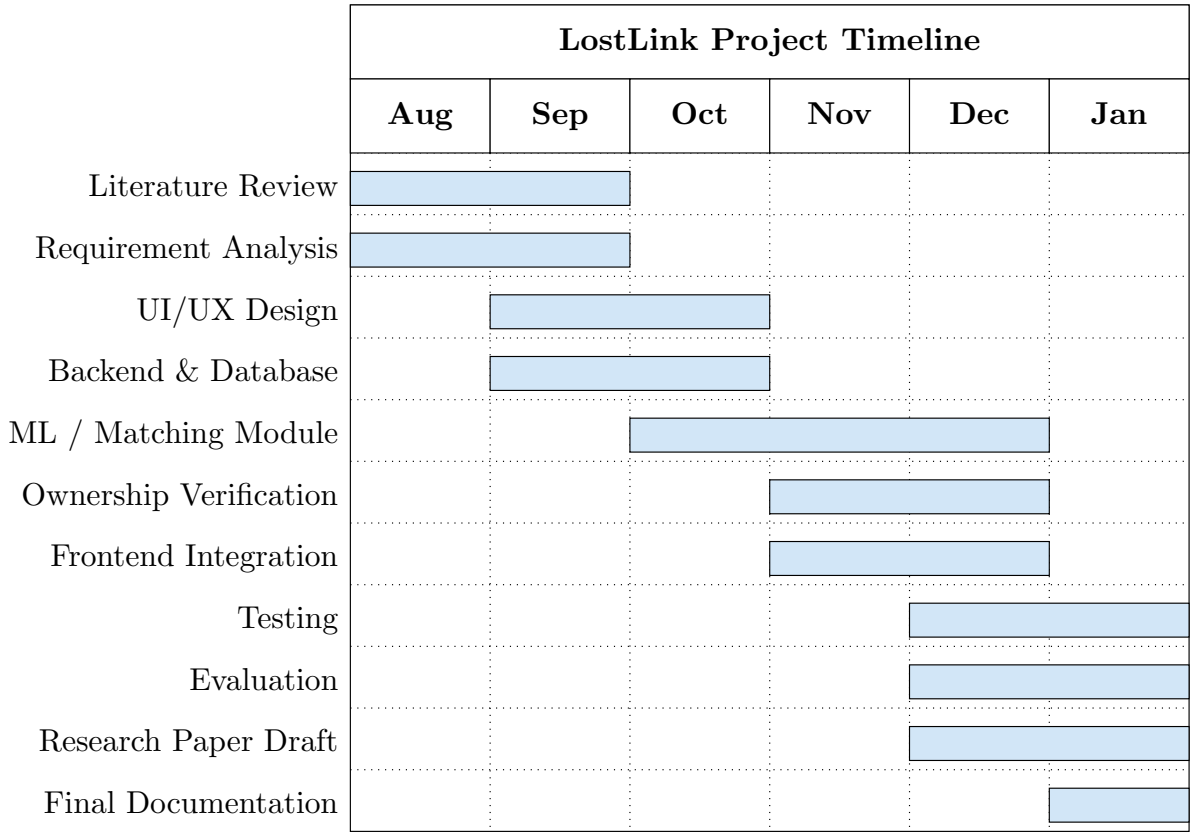


Figure 6.1: Project timeline (Gantt chart), August 2026 – January 2027.

Table 6.1: Project phases and their planned months

Phase	**Planned Months**
Literature Review	Aug – Sep
Requirement Analysis	Aug – Sep
UI/UX Design	Sep – Oct
Backend & Database	Sep – Oct
ML / Matching Module	Oct – Dec
Ownership Verification	Nov – Dec
Frontend Integration	Nov – Dec
Testing	Dec – Jan
Evaluation	Dec – Jan
Research Paper Draft	Dec – Jan
Final Documentation	Jan

Chapter 7

Conclusion

Numerous approaches have been used to study lost-and-found recovery, including keyword and metadata search, category-specific Siamese image matching, general-purpose zero-shot embeddings, multimodal fusion of image with text, location and time, and, least developed of all, ownership verification. While zero-shot vision-language and self-supervised models promise to overcome this limitation at the expense of being mainly untested for this particular application, convolutional Siamese networks provide strong accuracy but do not generalize beyond the category they were trained on. Ownership verification is still the least developed area; current systems either ignore it, replace it with a login check, or describe it conceptually without disclosing the false-acceptance and false-rejection rates that authentication research considers to be the typical method of assessing precisely this type of mechanism.

Both gaps are intended to be filled by LostLink. Both within a single item category and across categories, matching performance will be assessed using precision, recall, F1-score, and top-K retrieval accuracy, comparing CLIP, DINOv2, and a keyword/category baseline. False Acceptance Rate and False Rejection Rate at different thresholds will be used to assess ownership-verification performance, and identification-rate-at-rank will be used to assess multi-claimant ranking. The working target for the matching pipeline is 90–95% accuracy across multiple item categories, to be assessed once the system is built and tested rather than assumed in advance, based on accuracy figures gathered across the reviewed literature (roughly 75–97% depending on task and category breadth), none of which were measured on the specific, cross-category, near-duplicate matching problem LostLink addresses.

After the system has been put into use and assessed, future work will involve expanding the item-category set, testing the fine-tuning of the embedding models if a sufficiently large in-domain dataset becomes available, and investigating whether the explainable score breakdown displayed to users affects how administrators actually resolve disputed claims.

References

[1] B. M. P. Dhanawardhana et al., "Enhancing Lost and Found Systems with Multi-Modal Deep Learning: Integrating SBERT and Siamese Networks for Improved Semantic Matching," *Advances in Artificial Intelligence and Machine Learning*, vol. 5, no. 2, pp. 3736–3754, 2025.

[2] M. Zhou et al., "LostNet: A Smart Way for Lost and Find," *PLOS ONE*, 2024, arXiv:2301.02277.

[3] A. Radford et al., "Learning Transferable Visual Models From Natural Language Supervision," in *Proceedings of ICML*, 2021.

[4] M. Oquab et al., "DINOv2: Learning Robust Visual Features without Supervision," arXiv:2304.07193, 2023.

[5] T. Khandelwal and S. Poria, "Multimodal Embeddings for Image-Text Retrieval: A Survey," *ACM Computing Surveys*, 2022.

[6] "Smart Lost and Found System for Campus," Zenodo preprint, 2026.

[7] R. Wadbudhe, V. Ganesh, A. Patil, A. Meshram, A. Sayyed, A. Hedau, "Uni Find: Lost and Found Objects Management System for Campus using AI & Cloud Services," *International Journal of Latest Technology in Engineering, Management & Applied Science*, vol. 14, no. 10, pp. 134–139, 2025.

[8] "An Effective Lost and Found System in University Campus," *International Journal of Advanced Research*, vol. 8, no. 32, pp. 99–112, 2023.

[9] T. Bienvenu, "AI-Driven Lost and Found Management System for Institutions," Master's thesis, School of Computing and Engineering Sciences, Strathmore University, Nairobi, Kenya, 2025.

[10] G. Koch, R. Zemel, R. Salakhutdinov, "Siamese Neural Networks for One-shot Image Recognition," *ICML Deep Learning Workshop*, 2015.

[11] J. Prawira, T. R. D. Saputri, "Lost Item Identification Model Development Using Similarity Prediction Method with CNN ResNet Algorithm," *Journal of Autonomous Intelligence*, vol. 7, no. 2, 2023.

[12] "Progressive Multi-Subspace Fusion for Text-Image Matching," *Complex & Intelligent Systems*, 2025.

[13] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, L.-C. Chen, "MobileNetV2: Inverted Residuals and Linear Bottlenecks," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018.

[14] "Attacks and Countermeasures in Fingerprint Based Biometric Cryptosystems," arXiv:1304.7386, 2013.

[15] "EVI: Multilingual Spoken Dialogue Tasks and Dataset for Knowledge-Based Enrolment, Verification, and Identification," arXiv:2204.13496, 2022.

[16] H. Zhao, S. Peng, "Design and Implementation of the Lost-and-Found System Based on Amap API," in *Proceedings of the IEEE 9th International Conference on Software Engineering and Service Science (ICSESS)*, 2018.

[17] Q. Xue, H. Ma, C. Sun, "Exploratory Research on Blockchain-Based Lost and Found Platform," in *Proceedings of the 7th International Conference on Intelligent Informatics and Biomedical Science (ICIIBMS)*, 2022.

[18] M. Ye, S. Chen, C. Li, W.-S. Zheng, D. Crandall, B. Du, "Transformer for Object Re-Identification: A Survey," arXiv:2401.06960, 2024.

[19] "Zero-Shot Object Re-Identification in Egocentric Kitchen Videos via Multi-Stage SAM3 Feature Fusion," arXiv:2605.26383, 2026.

Appendix A

Plagiarism Report

A.1 Plagiarism Report

[To be inserted after the plagiarism-check tool provided by the institute has been run on the final report.]

A.2 AI Content Report

[To be inserted after the AI-content-detection tool provided by the institute has been run on the final report.]